

PCDD, PCDF, and PCB in farm-raised catfish from southeast United States--concentrations, sources, and CYP1A induction.



Nine catfish fillets, three catfish nuggets, two feed samples, and one pond sediment were analyzed for PCDD, PCDF, and PCB. Farm-raised catfish from Mississippi, Alabama, and Arkansas contained significant levels of 2,3,7,8-substituted PCDD and PCDF. In addition, a large number of non-2,3,7,8-substituted congeners were present in all samples. The catfish fillets and catfish nuggets also contained high concentrations of dioxin-like PCB, as well as a number of non-dioxin-like PCB. The TEQ based on PCDD and PCDF ranged from 9.5 to 43.0 pg/g lipid and the TEQ based on PCB ranged from 0.45 to 4.9 pg/g lipid for all catfish samples. The dioxin-like PCB contributed 4-16% to the total TEQ (PCDD/PCDF/PCB) for the catfish samples. The major source for the PCDD, PCDF, and PCB appears to be from feed and not from pond sediment. Immunoreactive CYP1A protein was elevated 2.5 fold in the pond-raised catfish compared to the aquarium-raised one. The results of this study suggest that the PCDD/PCDF are more important than the PCB in the CYP1A induction.